PRODUCTION AI ENGINEERING

120

REAL-WORLD SCENARIOS

Lead AI Engineer

Interview Question Bank

12 CATEGORIES • 120 QUESTIONS

Backend & APIs • LLMs • RAG • Agentic AI

DevOps • Kubernetes • Monitoring • Security

System Design • Leadership • FDE

Learn. Diagnose. Design. Lead.

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WHAT’S INSIDE

12 categories • 10 questions each

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Backend Development & APIs

10 REAL-WORLD INTERVIEW SCENARIOS

Q1 Your FastAPI application works locally but becomes slow when 500 concurrent users access it. How would you diagnose and resolve the bottleneck?

Q2 An API endpoint takes 12 seconds to respond, but the database query takes only 100 ms. How would you isolate the latency?

Q3 Your Python service occasionally runs out of memory while processing large PDFs. How would you investigate and fix it?

Q4 Multiple users upload documents simultaneously and some processing jobs fail. How would you redesign the workflow?

Q5 An API sometimes creates duplicate records when clients retry requests. How would you implement idempotency?

Q6 Your backend has 20 tightly coupled modules and every release breaks another feature. How would you refactor it?

Q7 An asynchronous FastAPI endpoint still blocks other requests. What would you inspect?

Q8 Your application needs to process a 1 GB document without exceeding pod memory limits. What architecture would you choose?

Q9 A critical dependency fails intermittently. How would you implement timeouts, retries, circuit breakers, and fallback behavior?

Q10 Two engineers disagree about microservices versus a modular monolith. How would you decide?

LLM & Generative AI Engineering

10 REAL-WORLD INTERVIEW SCENARIOS

Q1 Your LLM generates different answers for the same question. How would you improve consistency?

Q2 An LLM hallucinates policy clauses even though the correct documents are available. How would you troubleshoot?

Q3 Your prompt performs well on 90% of cases but fails on complex queries. How would you systematically improve it?

Q4 Your AI application spends INR 5 lakh per month on tokens. How would you reduce cost without significantly degrading quality?

Q5 GPT-based responses take 15 seconds. How would you optimize latency?

Q6 Your model produces invalid JSON in production. How would you enforce reliable structured outputs?

Q7 A new model version improves quality but doubles inference cost. How would you decide whether to migrate?

Q8 Users ask multilingual questions but the LLM performs poorly in some languages. What would you change?

Q9 A customer wants to fine-tune an LLM instead of using RAG. How would you evaluate the decision?

Q10 Your primary LLM provider becomes unavailable. How would you design model failover and graceful degradation?

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RAG, Search & Vector Databases

10 REAL-WORLD INTERVIEW SCENARIOS

Q1 Your RAG system retrieves irrelevant chunks even though relevant documents exist. How would you debug it?

Q2 A 500-page contract changes only on three pages. How would you implement incremental indexing?

Q3 Similarity search returns the correct chunk at rank 12, but your top-k is 5. How would you improve retrieval?

Q4 Your reranker improves relevance but adds 800 ms latency. How would you balance speed and quality?

Q5 Your pgvector database grows from 100,000 to 50 million embeddings. How would you redesign indexing and retrieval?

Q6 Users ask questions requiring data from multiple documents. How would you design multi-hop retrieval?

Q7 Your system retrieves correct content but produces incorrect citations. How would you fix citation grounding?

Q8 Your documents contain tables, scanned pages, images, and multilingual text. How would you design ingestion?

Q9 A user asks for a supply chain manager but the retrieved contract contains a placeholder instead of a real name. How should the system respond?

Q10 Your hybrid BM25 plus vector search underperforms vector-only search. How would you diagnose the fusion and ranking strategy?

Agentic AI & Multi-Agent Systems

10 REAL-WORLD INTERVIEW SCENARIOS

Q1 An AI agent repeatedly calls the same tool and enters an infinite loop. How would you prevent it?

Q2 Your multi-agent workflow is slower and more expensive than a single-agent approach. How would you evaluate whether multiple agents are justified?

Q3 An agent executes the wrong business action despite correctly understanding the user request. How would you design safer execution?

Q4 Your LangGraph workflow fails halfway through a five-step business process. How would you resume safely?

Q5 Two agents provide conflicting recommendations. How would you design validation and resolution?

Q6 Your agent has access to 50 tools and frequently selects the wrong one. How would you improve tool routing?

Q7 An MCP server becomes unavailable while an agent is executing a workflow. How would you handle it?

Q8 Your agent needs to retain user context across multiple sessions. How would you design short-term and long-term memory?

Q9 A customer asks for autonomous contract approval. What actions would require human approval?

Q10 How would you measure whether an agent successfully completes tasks rather than merely generating plausible responses?

DevOps, CI/CD & MLOps

10 REAL-WORLD INTERVIEW SCENARIOS

Q1 Your AI service works in DEV but fails in UAT. How would you systematically investigate environment drift?

Q2 A GitHub Actions pipeline succeeds but the deployed container crashes. What would you inspect?

Q3 You need to deploy a new prompt, model, and application version independently. How would you structure CI/CD?

Q4 Your Docker image is 5 GB and builds slowly. How would you optimize it?

Q5 A model update unexpectedly reduces production accuracy. How would you detect and roll back the change?

Q6 Your team frequently deploys broken configurations. How would you introduce automated validation?

Q7 You have separate DEV, UAT, and PROD Azure environments. How would you manage secrets and infrastructure configuration?

Q8 A critical vulnerability is discovered in a Python dependency. How would you remediate it safely?

Q9 Your AI pipeline requires reproducible embeddings, prompts, datasets, and model versions. How would you implement versioning?

Q10 How would you design an end-to-end CI/CD pipeline for FastAPI, LangGraph, pgvector, and AKS?

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Cloud Deployment & Kubernetes

10 REAL-WORLD INTERVIEW SCENARIOS

Q1 Your AKS pod continuously restarts with CrashLoopBackOff. How would you troubleshoot it?

Q2 Your AI API works through the Kubernetes service but fails through NGINX Ingress. What would you check?

Q3 Traffic increases from 100 to 10,000 requests per hour. How would you scale the application?

Q4 Your pods have high CPU utilization but low memory utilization. How would you configure autoscaling?

Q5 An AKS deployment causes 30 seconds of downtime. How would you implement zero-downtime deployment?

Q6 Your document-processing workers interfere with user-facing inference latency. How would you isolate workloads?

Q7 Your production deployment works but cannot connect to Azure OpenAI or PostgreSQL. How would you investigate networking?

Q8 Your service is deployed across regions and one Azure region fails. How would you design disaster recovery?

Q9 Your Kubernetes cluster costs increase by 60%. How would you identify and reduce waste?

Q10 How would you deploy a production RAG platform using ACR, AKS, Ingress, Azure Blob Storage, and PostgreSQL?

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Monitoring, Observability & Evaluation

10 REAL-WORLD INTERVIEW SCENARIOS

Q1 Users report incorrect AI answers, but your API success rate is 99.9%. How would you detect the actual problem?

Q2 Your RAG response latency suddenly increases from 3 seconds to 12 seconds. How would you identify the slow stage?

Q3 Your retrieval precision decreases after a document ingestion change. How would you detect and isolate the regression?

Q4 An agent uses excessive tokens and repeatedly invokes tools. What telemetry would you capture?

Q5 How would you design distributed tracing across FastAPI, retrieval, reranking, LLM inference, and tools?

Q6 Your evaluation dataset scores 95%, but customer satisfaction remains low. How would you investigate the mismatch?

Q7 How would you detect hallucinations, unsupported claims, and citation errors in production?

Q8 Your team receives hundreds of monitoring alerts daily. How would you reduce alert fatigue?

Q9 Your LLM provider reports successful requests, but users receive incomplete answers. What application-level metrics would you introduce?

Q10 How would you design an AI observability dashboard for quality, latency, cost, reliability, and security?

Security, Governance & Responsible AI

10 REAL-WORLD INTERVIEW SCENARIOS

Q1 User A receives a contract excerpt belonging to User B. How would you contain, investigate, and permanently prevent this incident?

Q2 A malicious document instructs your RAG assistant to ignore system rules. How would you defend against prompt injection?

Q3 Your customer wants to ensure confidential documents never leave an approved region. How would you design the solution?

Q4 Your AI application must support tenant isolation across 100 enterprise customers. How would you enforce it?

Q5 An attacker attempts to bypass authorization by manipulating document metadata filters. How would you secure retrieval?

Q6 Your AI system processes personal data and customers request deletion. How would you handle data across blobs, chunks, embeddings, caches, and logs?

Q7 Your agent can send emails and update financial records. How would you manage permissions and approval workflows?

Q8 An API key is accidentally committed to GitHub. What immediate and long-term actions would you take?

Q9 Your model gives biased recommendations for certain customer groups. How would you evaluate and mitigate the problem?

Q10 How would you design end-to-end AI governance covering access control, auditability, evaluations, safety, and compliance?

Data Engineering & Database Design

10 REAL-WORLD INTERVIEW SCENARIOS

Q1 Your ingestion pipeline processes 10,000 PDFs daily, but a backlog develops. How would you scale it?

Q2 Your PostgreSQL query latency increases sharply after embedding growth. How would you investigate?

Q3 An upstream source sends duplicate and out-of-order events. How would you ensure correct processing?

Q4 Your ETL pipeline fails after processing 80% of documents. How would you support checkpointing and recovery?

Q5 Your application needs both structured SQL queries and semantic retrieval. How would you design hybrid access?

Q6 Your ingestion pipeline must handle document updates and deletions while preserving query consistency. How would you design it?

Q7 Your customer needs near-real-time knowledge updates. How would you move from batch ingestion to event-driven processing?

Q8 How would you select between PostgreSQL pgvector, a managed vector database, and a search engine?

Q9 Your embedding model changes, making existing vectors incompatible. How would you migrate without downtime?

Q10 Your analytics team wants to use operational AI traces for reporting. How would you design the data pipeline?

System Design, Scalability & Cost

10 REAL-WORLD INTERVIEW SCENARIOS

Q1 Design an enterprise document intelligence platform supporting 10,000 users and millions of documents.

Q2 Your customer requires a p95 response latency below 3 seconds. How would you allocate the latency budget?

Q3 Your AI solution must support 99.9% availability. How would you design reliability and recovery?

Q4 Your customer needs an on-premises deployment because cloud-hosted models are prohibited. What architecture would you propose?

Q5 You have an INR 10 lakh monthly infrastructure budget. How would you estimate and optimize TCO?

Q6 How would you design a multi-region, multi-tenant RAG service with strong tenant isolation?

Q7 Your architecture needs to support 100x traffic growth over two years. Which components would you scale independently?

Q8 Your customer asks for a real-time AI assistant combining SQL, documents, and external APIs. How would you orchestrate the workflow?

Q9 Your system must process 1 million documents while keeping ingestion and inference workloads isolated. How would you design it?

Q10 How would you decide between synchronous APIs, background queues, event-driven services, and workflow orchestration?

Engineering Leadership & Incident Management

10 REAL-WORLD INTERVIEW SCENARIOS

Q1 Two senior engineers strongly disagree about the AI architecture. How would you resolve the disagreement?

Q2 Your team repeatedly misses delivery deadlines. How would you diagnose and improve execution?

Q3 A production incident occurs at midnight and affects a major customer. How would you lead the response?

Q4 A junior engineer introduces a bug that exposes confidential data. How would you handle the incident and the engineer?

Q5 Your team wants to adopt a new agent framework, but the current solution is stable. How would you evaluate the change?

Q6 Management wants delivery in four weeks, but engineering estimates eight. How would you negotiate scope and risk?

Q7 How would you establish code review, testing, design review, and deployment standards for an AI engineering team?

Q8 Your team has strong ML engineers but limited production engineering experience. How would you close the gap?

Q9 Your AI system has accumulated technical debt while feature requests keep increasing. How would you prioritize?

Q10 How would you communicate an AI quality regression to executives and customers?

Customer, Business & FDE Scenarios

10 REAL-WORLD INTERVIEW SCENARIOS

Q1 A customer says, 'We want an AI agent,' but cannot define the business problem. How would you conduct discovery?

Q2 Your customer wants ChatGPT-like capabilities over confidential enterprise documents. How would you propose a solution?

Q3 An AI proof of concept works well, but users refuse to adopt it. How would you investigate?

Q4 Your customer expects 100% accurate answers from an LLM. How would you establish realistic acceptance criteria?

Q5 A customer demands a custom fine-tuned model. How would you compare alternatives and justify your recommendation?

Q6 Your customer says your AI solution is too expensive. How would you demonstrate ROI and propose cost reductions?

Q7 An executive asks why the company cannot simply upload PDFs to a chatbot instead of building enterprise RAG. How would you respond?

Q8 Your AI pilot improves productivity by 30%, but the customer questions the measurement. How would you validate business impact?

Q9 A critical customer requests an unsafe autonomous capability. How would you propose a safer alternative?

Q10 Your customer wants production deployment in 30 days. How would you define the MVP, milestones, risks, and success criteria?